

Precalc Lesson 1.10 Homework

Answer Key

1. No, $B(t)$ is not one-to-one because temperatures could be repeated throughout the day. This means that given a certain temperature, you cannot pinpoint what time of day it is.
2. No, f is not one-to-one because it has repeated outputs. This means the inverse relationship would have one input going to more than one output. This can be seen using a horizontal line test.
3. $y = x^2 - 3$
Domain: $x \geq 0$
Range: $y \geq -3$
4. Domain: $(-\infty, 0) \cup (0, \infty)$
Range: $(-\infty, -5) \cup (-5, \infty)$
Yes, the domain of f is the range of f^{-1} and the range of f is the domain of f^{-1} .
5. a. 6
b. $y = -5$
6. Domain: $(-\infty, 4) \cup (4, \infty)$ or $\{x \in \mathbb{R} | x \neq 4\}$
Range: $(-\infty, 2) \cup (2, \infty)$ or $\{y \in \mathbb{R} | y \neq 2\}$

7. You could restrict the domain to $x \leq 4$ or you can restrict the domain to $x \geq 4$ so you only have half the parabola and thus no y -values on f repeat.

8.

